

Recuva/CCleaner Assignment

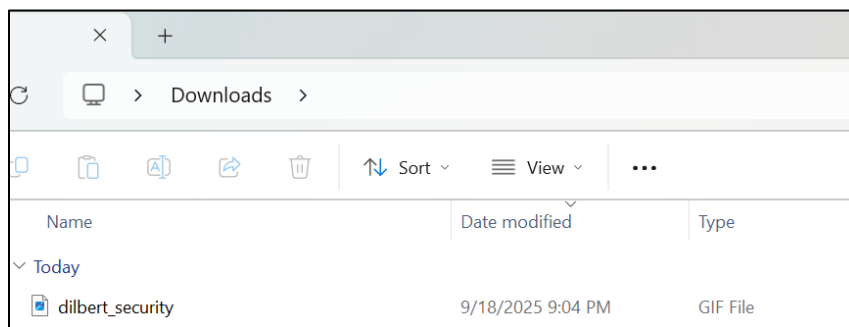
Recuva

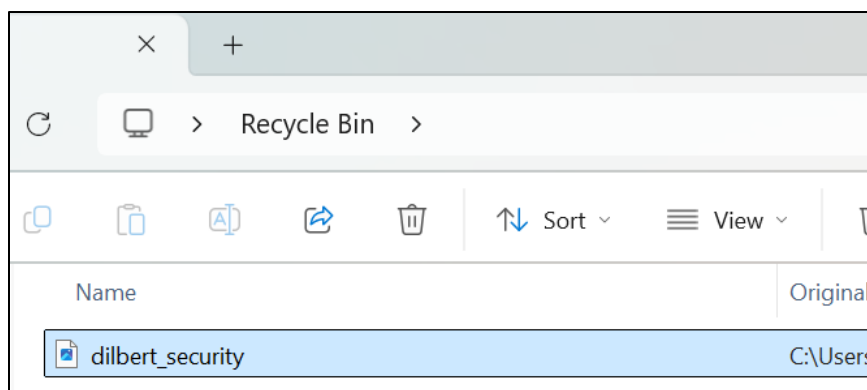
Question 1: what did I do?

For this assignment, I conducted an experiment to test how feasible it is to recover data that has been deleted. When a file is saved, the operating system (OS) does not simply place it in a single location on the disk. Instead, the OS relies on a file system to manage storage. Common file systems include NTFS (Windows), APFS/HFS+ (macOS), and ext4 (Linux). The file is broken down into blocks or clusters, which the OS writes into available sectors on the disk—these sectors may or may not be next to each other. The OS then records the file’s metadata (such as its name, type, size, timestamps, and physical block locations) in structures like the Master File Table (MFT) in NTFS or inode tables in ext4.

When a file is deleted, it is usually first moved to a temporary holding area such as the Recycle Bin in Windows or the Trash in macOS. At this stage, the file is not truly gone and can be easily restored. However, when the Recycle Bin or Trash is “permanently deleted”, the OS does not erase the actual data; rather, it removes the pointer to the file in the file system’s directory structure (e.g., the MFT or inode table) and marks the sectors previously occupied by the file as available. The data itself (the 1s and 0s) remains on the disk until those sectors are overwritten by new information. This means that “permanently deleted” files can still be recovered as long as their data blocks remain intact.

Data recovery tools, such as Recuva, exploit this property. They scan the disk for files that no longer have active directory entries but whose data has not yet been overwritten. If the space has not been reused, recovery is often straightforward; if it has been partially overwritten, recovery is less certain and may result in corrupted or incomplete files. For my experiment, I deliberately created a file containing test data (dilbert_security.gif), deleted it by moving it to the Recycle Bin, and then emptied the Recycle Bin. Afterwards, I attempted to recover the file using Recuva to see whether the tool could locate and restore it.





Question 2: what were the results?

Unfortunately, I was not able to restore the file despite using the deep scan feature, which may have been due to the limitations of the free version of the application, the possibility that the deleted sectors had already been overwritten, or other technical constraints such as file corruption. Under normal circumstances, Recuva begins with a quick scan that leverages file system entries, like the Master File Table (MFT) in NTFS, to locate references to recently deleted files that have not yet been overwritten. When these pointers are still intact, recovery can be relatively straightforward. However, if those references are lost or the file system has already reallocated the space, Recuva can conduct a deeper, sector-by-sector scan. This process attempts to reconstruct files by searching for recognizable file signatures and patterns across the raw disk surface. While this method is more thorough, its success depends heavily on the extent of overwriting and the file type involved. As a result, even though deep scans increase the chances of recovery, they cannot guarantee success in every case, especially when dealing with sensitive or fragmented data.

Recuva						
Recuva v1.54.120 (64-bit) Windows 11 Pro 64-bit (Admin) Intel Core i5-8269U CPU @ 1.60GHz, 16.0GB RAM, Intel UHD Graphics 620						
Select the files you want to Recover by ticking the boxes and then pressing Recover. For the best results, restore the files to a different drive.						
☐	Filename	Path	Last Modified	Size	State	Comment
☐	Preferences~RF50aaf5ef.TMP	C:\Users...	9/18/2025 22:11	108 KB	Excellent	No overwritten clusters detected.
☐	Preferences~RF50aa80af.TMP	C:\Users...	9/18/2025 22:11	108 KB	Unrecove...	This file is overwritten with "C:\Users\akeem\Ap...
☐	ccba5a5986c77e43.customDestinations-ms~RF50a8bad4.TMP	C:\Users...	9/18/2025 22:09	21 KB	Poor	This file is overwritten with "C:\ProgramData\H...
☐	Preferences~RF50a8c8de.TMP	C:\Users...	9/18/2025 22:09	108 KB	Unrecove...	File header data is empty or the file is securely ...
☐	RevisitationBloomfilter~RF50a90c01.TMP	C:\Users...	9/18/2025 22:09	5 KB	Unrecove...	File header data is empty or the file is securely ...
☐	Network Persistent State~RF50a98383.TMP	C:\Users...	9/18/2025 22:09	80 KB	Unrecove...	File header data is empty or the file is securely ...
☐	Network Persistent State~RF50aa9fe0.TMP	C:\Users...	9/18/2025 22:09	5 KB	Unrecove...	File header data is empty or the file is securely ...
☐	the-real-index~RF50a8c2d3.TMP	C:\Users...	9/18/2025 22:08	86 KB	Unrecove...	File header data is empty or the file is securely ...
☐	a3c9c4b5-2434-47a9-9b16-620c97f15356.tmp	C:\Users...	9/18/2025 22:06	970 KB	Poor	This file is overwritten with "C:\Windows\Prefet...
☐	the-real-index~RF50a8fef1.TMP	C:\Users...	9/18/2025 22:06	120 b...	Excellent	No overwritten clusters detected.
☐	001800000000D894778A8945	C:\\$Ext...	9/15/2025 22:06	81 KB	Unrecove...	File header data is empty or the file is securely ...
☐	000436.log	C:\Users...	9/18/2025 22:05	196 KB	Very poor	This file is overwritten with "C:\Users\akeem\Ap...
☐	Local State~RF50a4f2c0.TMP	C:\Users...	9/15/2025 22:31	16 KB	Unrecove...	File header data is empty or the file is securely ...
☐	000773.log	C:\Users...	9/15/2025 22:31	688 KB	Unrecove...	This file is overwritten with "C:\Users\akeem\Ap...
☐	wdgts_conf[1].json	C:\Users...	9/15/2025 22:31	467 b...	Excellent	No overwritten clusters detected.
☐	TransportSecurity~RF50a4cc8a.TMP	C:\Users...	9/15/2025 22:31	356 b...	Excellent	No overwritten clusters detected.
☐	prefs.json~RF4153d1ca.TMP	C:\Progr...	9/15/2025 22:03	2 KB	Unrecove...	This file is overwritten with "C:\Users\akeem\Ap...
☐	Network Persistent State~RF50a5181a.TMP	C:\Users...	9/15/2025 22:00	5 KB	Unrecove...	This file is overwritten with "C:\Users\akeem\Ap...
☐	000437.ldb	C:\Users...	9/15/2025 21:59	43 KB	Unrecove...	This file is overwritten with "C:\Users\akeem\Ap...

Question 3: What did I learn?

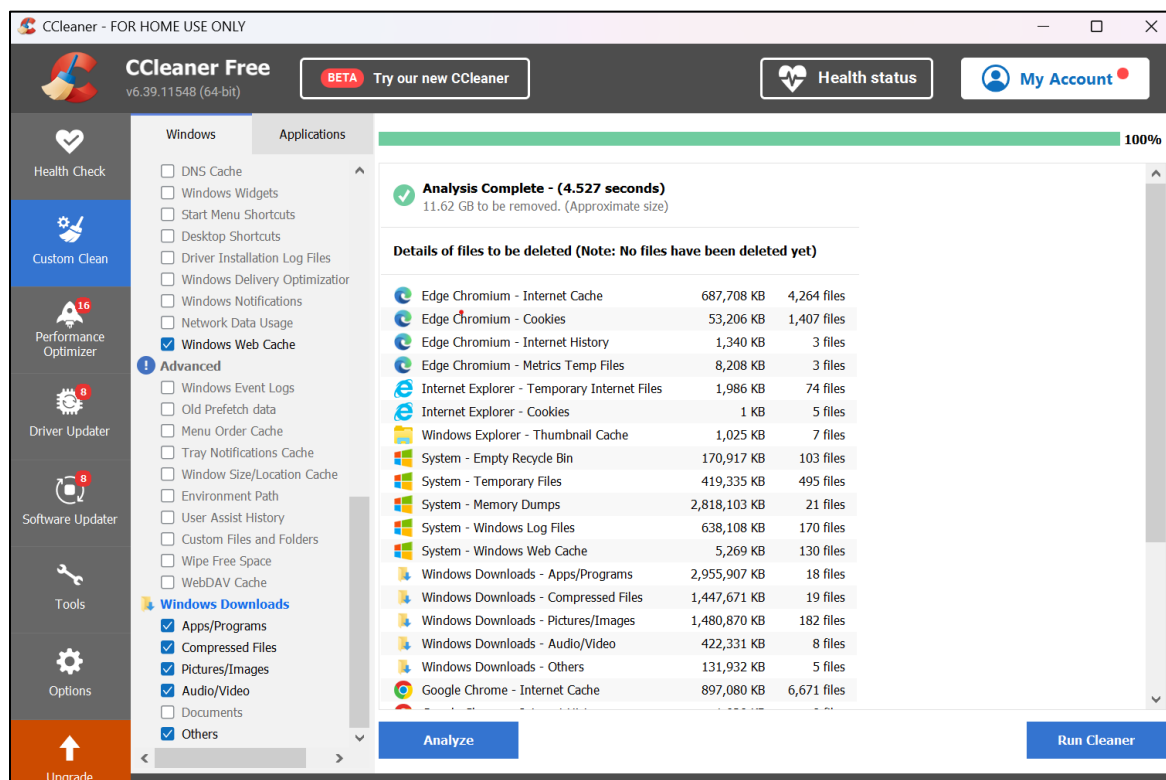
This exercise demonstrates that deleted files are not truly irrecoverable unless deliberate measures are taken. The fact that a free tool like Recuva can recover “permanently deleted” data underscores the risks of relying solely on standard deletion methods. While such recovery tools are valuable for restoring accidentally deleted files, they also reveal how easily malicious actors could exploit residual data left behind on a drive.

From a cybersecurity standpoint, the lesson is clear: secure deletion practices are critical. Conventional file deletion merely removes directory references, leaving the underlying data blocks intact and vulnerable. Without techniques such as overwriting, secure wiping, or encryption, sensitive information remains recoverable and could expose individuals or organizations to data breaches, identity theft, or regulatory violations. Therefore, implementing reliable secure deletion methods is essential to ensure that confidential data is truly destroyed and cannot be reconstructed.

CCleaner

Question 1: what did I do?

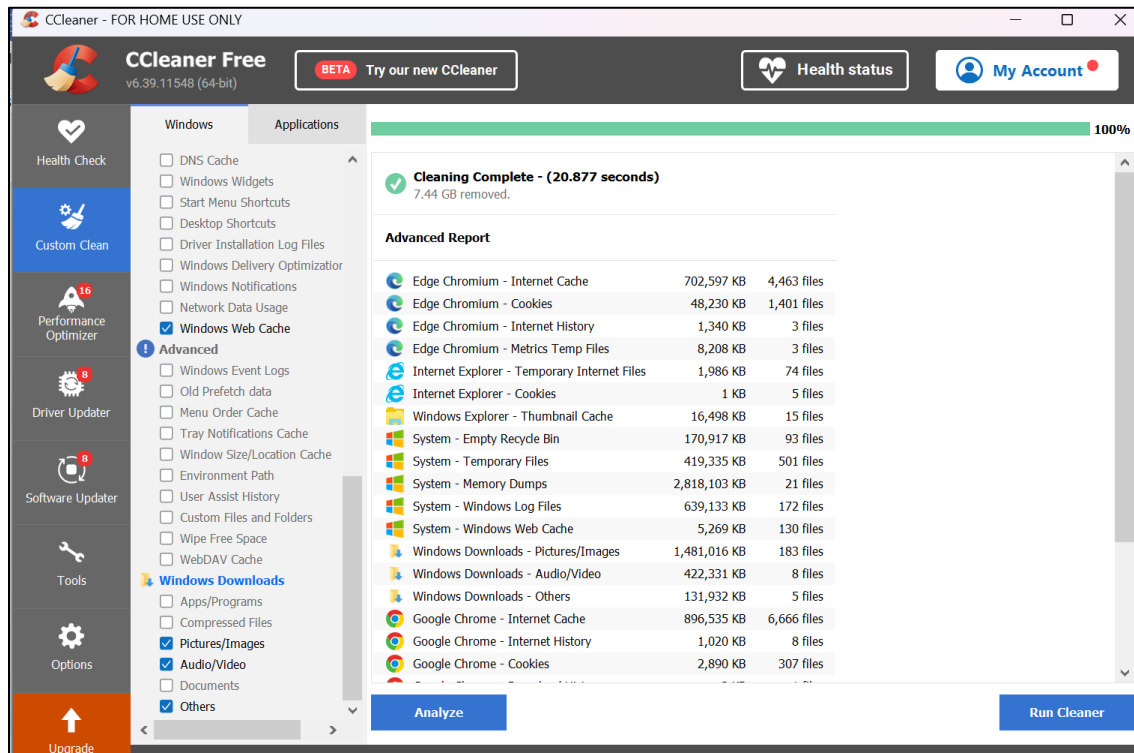
I used CCleaner to illustrate a contrasting process. Unlike standard file deletion, which only removes directory pointers, CCleaner is specifically designed to permanently erase files by overwriting the disk space they once occupied. This overwriting process makes recovery virtually impossible. Users can even select different levels of overwriting, which increases security by ensuring that deleted data cannot be reconstructed. The overwriting itself can be performed with random data or specific binary patterns (such as 00000000, 11111111, or random byte sequences). When I ran CCleaner, the application first analyzed the target folders and provided a report on how much storage space would be reclaimed. After authorizing the cleanup, CCleaner successfully erased the selected data and freed a substantial amount of space on my system.

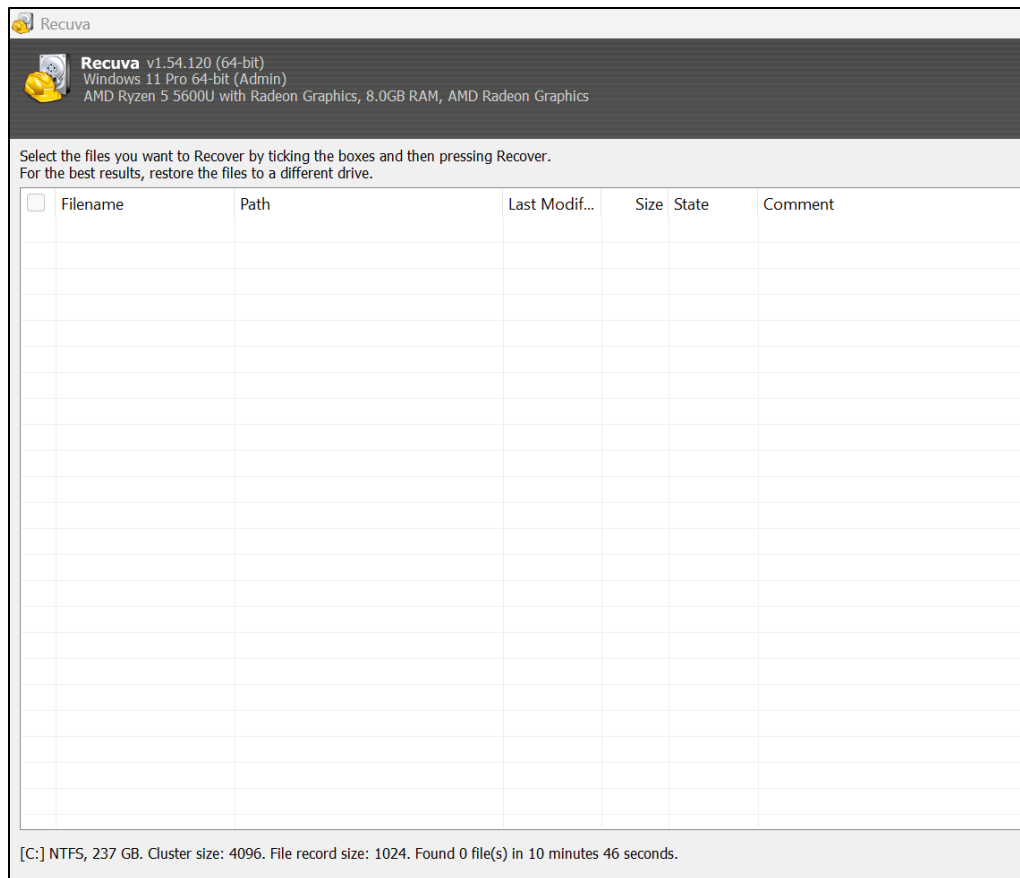


Question 2: what were the results?

The results were astonishing! Not only did the CCleaner application completely clear out my Recycle Bin, but it also freed up a significant amount of storage space on my computer. In fact, the persistent “storage almost full” warning I had been receiving on my personal computer disappeared after the cleaning process. This demonstrated not only the tool’s effectiveness in securely deleting files but also its ability to optimize overall system performance by reclaiming wasted space. After the cleanup, I attempted to run the Recuva application to test whether I could recover any of the deleted Recycle Bin items. To my surprise, nothing was detected—no

traces of the files appeared in the scan results, even after a deep scan. This strongly suggests that CCleaner had not just removed the files but had actively overwritten the sectors where the files once resided, rendering them unrecoverable.





Question 3: What did I learn?

CCleaner demonstrates the importance of securely overwriting files to achieve true data sanitization. Standard deletion only removes file references, leaving the underlying data intact and vulnerable to recovery. By contrast, CCleaner performs secure erasure, overwriting disk space so that even advanced recovery tools cannot reconstruct the information. This exercise highlighted how improper disposal of digital files (like HR records, financial data, or personal information) can lead to data breaches or identity theft if not securely erased. It also showed why organizations mandate strict data disposal policies, such as disk wiping or encryption, to protect confidentiality and maintain trust. In conclusion, this exercise reinforced the importance of using secure disposal methods and organizational policies to prevent data breaches, and protect confidentiality.